EXNO: 1	**Estimation of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line from Primary Constants**
DATE:	

Test Case 1:

Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line using the following parameters

a) Resistance =10 Ω/Km

b) Capacitance = 8 µF/Km

c) Inductance =3.5 mH/Km

d) Conductance =4 µmho/Km

e) Frequency= 2 Khz

AIM: Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line with Resistance =10 Ω/Km, Capacitance = 8 µF/Km, Inductance =3.5 mH/Km, Conductance =4 µmho/Km, Frequency= 2 Khz

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

CALCULATION:

PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
```

```
 Y=G+(%i*w*C);
 Zo=sqrt(Z/Y);
=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

```
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

RESULT:
The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

Test Case 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L, G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

AIM: To examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L,G and C values constant and calculate the Characteristic

Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

CALCULATION:

PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:

Variable R
Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.No	R	Zo(ohms)	Alpha	Beta
1	5	13.72 + j(-0.91)	0.1822	2.7592
2	10	13.81 + j(-1.80)	0.3621	2.7769
3	15	13.95 + j(-2.67)	0.5376	2.8051
4	20	14.14 + j(-3.52)	0.7074	2.8426
5	25	14.36 + j(-4.33)	0.8704	2.8875

Variable L

Constants R=10Ω; f=2000; G=.4*(10^-6); C=8*(10^-6).

RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is examined at a frequency f=2.5 Khz.

Test Case 3:

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L, R, G and C values constant and calculate the Characteristic Impedance and Propagation Constant. Vary the capacitance values from 8 µF to 16 µF at the interval of 2 µF with R,G and L values constant.

AIM: To explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L, R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant

CALCULATION:

PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
```

```
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:

Variable f

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.No	f(Hz)	Zo(ohms)		Alpha
1	1000	19.99+j(-4.97)	0.2501	1.0050
2	1500	19.66+j(-3.37)	0.2544	1.4821
3	2000	19.53+j(-2.55)	0.2560	1.9635
4	2500	19.47+j(-2.04)	0.2568	2.4470
5	3000	19.44+j(-1.71)	0.2572	2.9315

-->

Variable C

Constants R=10; L=.009; f=2500; G=.4*(10^-6);

S.No	C(F)	Zo(ohms)		Alpha
1	4.000000e-06	47.46+j(-1.68)	0.1054	2.9822
2	8.000000e-06	33.56+j(-1.18)	0.1490	4.2175
3	1.200000e-05	27.40+j(-0.97)	0.1825	5.1654
4	1.600000e-05	23.73+j(-0.84)	0.2107	5.9645
5	2.000000e-05	21.23+j(-0.75)	0.2356	6.6685

;

RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is investigated for various values of frequencies and capacitance values.